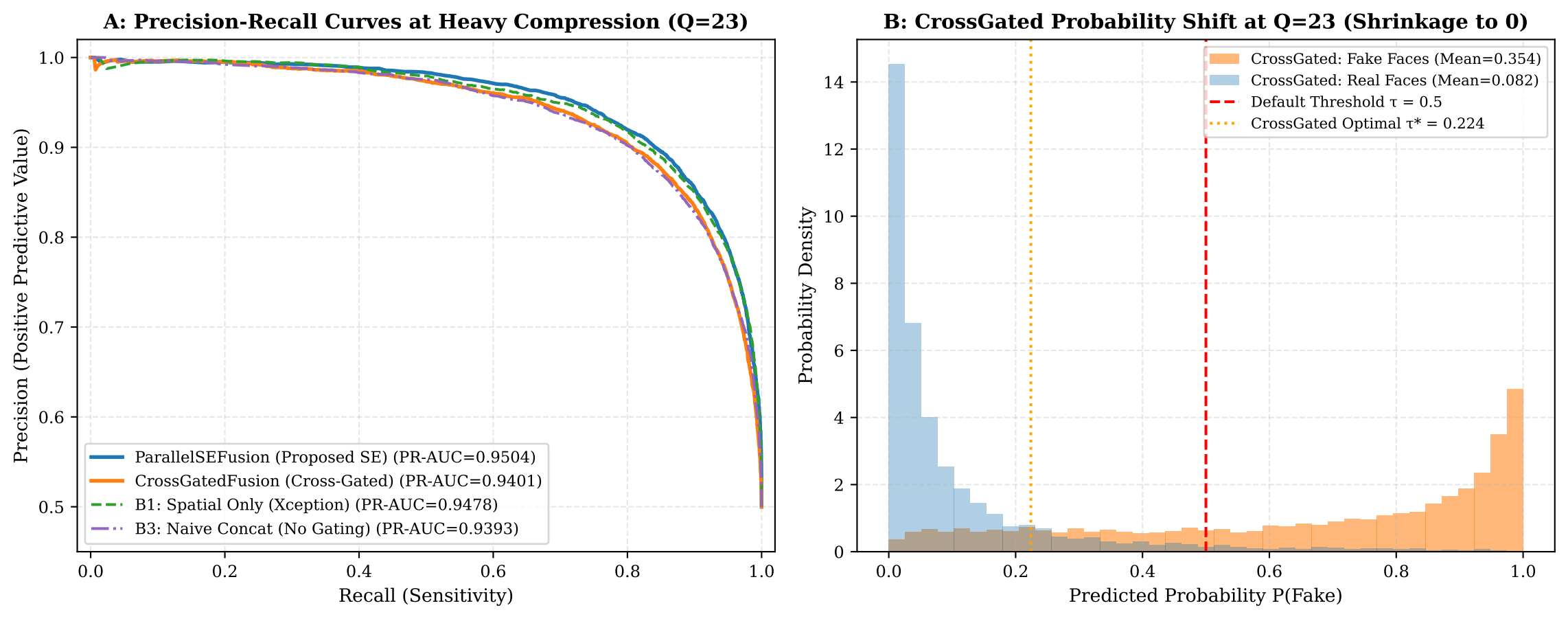

A: Precision-Recall Curves at Heavy Compression (Q=23)
Precision (Positive Predictive Value)
Recall (Sensitivity)
ParallelSEFusion (Proposed SE) (PR-AUC=0.9504)
CrossGatedFusion (Cross-Gated) (PR-AUC=0.9401)
B1: Spatial Only (Xception) (PR-AUC=0.9478)
B3: Naive Concat (No Gating) (PR-AUC=0.9393)
B: CrossGated Probability Shift at Q=23 (Shrinkage to 0)
Probability Density
Predicted Probability P(Fake)
CrossGated: Fake Faces (Mean=0.354)
CrossGated: Real Faces (Mean=0.082)
Default Threshold τ = 0.5
CrossGated Optimal τ* = 0.224